

-- Create schema

```
CREATE SCHEMA IF NOT EXISTS shopey;
```

-- Customers

```
CREATE TABLE shopey.customers (  
  customer_id SERIAL PRIMARY KEY,  
  first_name VARCHAR(100) NOT NULL,  
  last_name VARCHAR(100) NOT NULL,  
  email VARCHAR(255) UNIQUE NOT NULL,  
  phone VARCHAR(20),  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

-- Categories

```
CREATE TABLE shopey.categories (  
  category_id SERIAL PRIMARY KEY,  
  category_name VARCHAR(100) NOT NULL,  
  description TEXT  
);
```

-- Vendors

```
CREATE TABLE shopey.vendors (  
  vendor_id SERIAL PRIMARY KEY,  
  vendor_name VARCHAR(150) NOT NULL,  
  contact_email VARCHAR(255)  
);
```

-- Products

```
CREATE TABLE shopey.products (  
  product_id SERIAL PRIMARY KEY,  
  product_name VARCHAR(200) NOT NULL,  
  category_id INT REFERENCES shopey.categories(category_id),  
  vendor_id INT REFERENCES shopey.vendors(vendor_id),  
  unit_price NUMERIC(10,2) NOT NULL CHECK (unit_price >= 0),  
  stock_qty INT DEFAULT 0  
);
```

-- Orders

```
CREATE TABLE shopey.orders (  

```

```
order_id    SERIAL PRIMARY KEY,
customer_id INT NOT NULL REFERENCES shopecustomers(customer_id),
order_date  TIMESTAMP DEFAULT NOW(),
status      VARCHAR(20) CHECK (status IN ('Pending','Confirmed','Shipped','Delivered','Cancelled'))
DEFAULT 'Pending'
);
```

-- Order Lines

```
CREATE TABLE shopec.order_lines (
  line_id    SERIAL PRIMARY KEY,
  order_id   INT NOT NULL REFERENCES shopec.orders(order_id),
  product_id INT NOT NULL REFERENCES shopec.products(product_id),
  quantity   INT NOT NULL CHECK (quantity > 0),
  unit_price NUMERIC(10,2) NOT NULL
);
```

-- Payments

```
CREATE TABLE shopec.payments (
  payment_id SERIAL PRIMARY KEY,
  order_id   INT UNIQUE NOT NULL REFERENCES shopec.orders(order_id),
  payment_date TIMESTAMP,
  method      VARCHAR(50) CHECK (method IN ('Card','PayPal','Bank Transfer','Wallet')),
  amount      NUMERIC(10,2) NOT NULL,
  status      VARCHAR(20) CHECK (status IN ('Pending','Paid','Failed','Refunded')) DEFAULT 'Pending'
);
```

10-Minute Coding Activity

Duration: 10 minutes (strictly timed) | Format: Individual | Tools: Any SQL editor / PostgreSQL or Azure SQL

Objective: Write and execute a working SQL query that meets the specification below. No AI assistance.

The Challenge

Using the ShopEY database schema (from Lab 1), write a SINGLE SQL query that returns the following result set:

customer_name	total_orders	total_spent	customer_rank
Alice Smith	8	£1,240.50	1
Bob Johnson	6	£980.00	2
Carol White	5	£870.25	3
...	...	...	...

Requirements

- customer_name: concatenated first_name + ' ' + last_name
- total_orders: count of distinct orders placed by each customer
- total_spent: sum of (quantity × unit_price) across all order lines for each customer
- customer_rank: rank by total_spent descending — use RANK() window function
- Include ONLY customers who have placed at least 1 order
- Order the result by customer_rank ascending